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Personal Color Analysis with Vision-Language Models

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ABSTRACT Recent advances in Vision-Language Models (VLMs) have shown remarkable success in multimodal understanding and reasoning across diverse domains. However, their application on personal color analysis remains largely underexplored. Personal color analysis aims to identify color palettes that harmonize with an individual's natural appearance, providing guidance for fashion, cosmetics, and styling. Traditional analysis relies heavily on in-person expert consultation and diagnosis, which restricts its widespread accessibility and practical use for the general public. In this paper, we propose Personal Color Vision-Language Models (PC-VLMs), a domain-specific multimodal framework for automated personal color analysis. We construct PCVLM-DB, the first expert-level vision-language dataset for personal color analysis, consisting of approximately 8,700 facial images and 30,000 question-answer pairs with diagnostic rationales. Leveraging this dataset, we perform supervised fine-tuning on the various pre-trained vision-language models to align visual representations with expert-level personal color knowledge. Our experimental results demonstrate that PC-VLMs significantly outperform both conventional vision-only classifiers and proprietary zero-shot multimodal models in personal color classification, while also providing interpretable reasoning for its predictions. The best model achieves 91.13% accuracy for warm/cool and 85.94% for seasonal tone classification, as well as 70.0% for fine-grained subtype prediction. It also produces more balanced predictions across subtype classes, reducing inter-class performance variation from over 30% to 14.78%. These results demonstrate the potential of vision-language models for perceptual and aesthetic decision-making tasks and provide a foundation for future multimodal systems in fashion, beauty, and personalized styling applications.

INDEX TERMS Vision–Language Models, Personal Color Analysis, Multimodal Learning, Facial Image Analysis, Cosmetics, Beauty.

I. INTRODUCTION

PERSONAL color analysis, also known as seasonal color analysis or armocromia, is an approach to identifying color palettes that harmonize with an individual's unique physical attributes, such as skin undertone, hair color, and iris color patterns [1]–[3]. By determining the optimal color category, individuals can enhance their visual appearance in fashion, cosmetics, and styling. Traditionally, this diagnostic process relies on professional consultants who manually apply physical color drapes based on established color theories, such as the Munsell system and the Practical Color Coordinate System (PCCS) [3]–[6]. However, these approaches are inherently subjective, time-consuming, and limited in accessibility, as expert-level analysis is typically available only through in-person consultations. To address these limitations, several automated diagnosis tools have been introduced, leveraging rule-based or computer vision methods to improve accessibility [7]–[9]. These approaches typically estimate personal color attributes by analyzing facial color distributions within specific color spaces, or employing standard image classification models for high-level tone categorization. While these approaches have demonstrated the feasibility of automated personal color analysis, they often struggle with fine-grained subtype estimation and fail to provide professional-level diagnostic reasoning. Consequently, their limited ability to capture subtle color variations and complex aesthetic cues reduces their reliability and interpretability in real-world scenarios.

Recent advancements in Vision-Language Models (VLMs) have shown promising results across diverse multimodal tasks

and specialized domains by effectively aligning visual representations with textual descriptions. In particular, multimodal foundation models such as ChatGPT [10], [11], Gemini [12], [13], and Qwen-VL [14], [15] have seen broad adoption in real-world applications and have also been actively investigated for domain-specific research, including autonomous-driving, robotics, and medical image understanding [16]–[19]. However, despite these impressive achievements and strong generalization capabilities, these models are primarily designed for general-purpose tasks and lack domain-specific understanding required for professional-level color analysis and aesthetic harmony assessment. As a result, their application in personal color analysis remains underexplored.

To bridge this gap, we propose a Personal Color Vision-Language Model (PC-VLM), a domain-specific model tailored for personal color analysis grounded in expert chromatic knowledge, enabling precise classification and professional reasoning. We construct a specialized VQA dataset named PCVLM-DB, by integrating approximately 8,700 high-quality facial images with around 30,000 single-turn textual conversations. PCVLM-DB is systematically annotated with a three-level hierarchical labeling scheme, covering warm/cool binary types, four seasonal categories, and eight fine-grained subtypes. Leveraging this dataset, we fine-tuned multiple vision-language foundation models to align visual representations with expert-level chromatic descriptions and hierarchical personal color taxonomy. Consequently, PC-VLM generates hierarchical personal color predictions and professional textual rationales, providing accurate classification with expert-level reasoning.

Our main contributions are summarized as follows:

1) We construct PCVLM-DB, the first VQA dataset for personal color analysis. It integrates approximately 8,700 high-quality images with 30,000 question-answering pairs, organized into a three-level hierarchical taxonomy.
2) We show that PC-VLM outperforms supervised vision classifiers and zero-shot commercial multimodal foundation models in personal color classification.
3) We demonstrate the capability of PC-VLM to produce structured textual reasoning aligned with professional standards, providing human-interpretable evidence that supports its diagnostic judgment.

II. RELATED WORKS

A. PERSONAL COLOR

Personal Color Theory. Jackson, Carole [1] introduced the four-season color theory, classifying individuals into spring, summer, autumn, and winter types based on their natural undertones. This framework has been widely adopted in the fashion and beauty industries as a fundamental guideline for recommending color palettes and products tailored to individual characteristics. Spring and autumn are generally associated with warm, yellowish undertones, with spring favoring light and clear colors and autumn aligning with rich, earthy tones. Conversely, summer and winter exhibit cooler, pinkish undertones. Summer is suited to soft, muted colors, whereas winter is characterized by vivid, intense hues. This seasonal classification provides an intuitive categorization of personal color impressions.

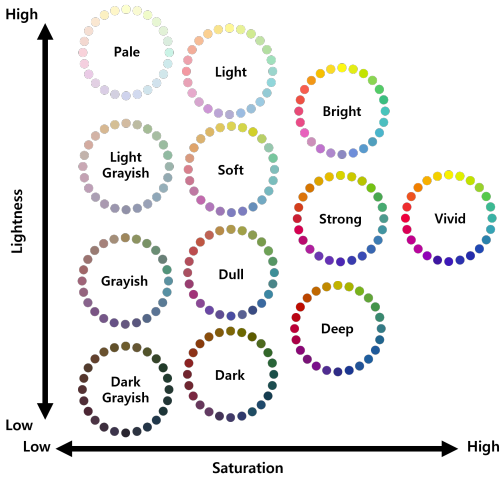


FIGURE 1. The structure of the Practical Color Coordinate System (PCCS). The system consists of a 24-hue color wheel and a tone chart, where tone is defined by the combination of lightness and saturation. PCCS provides 12 distinct tonal classes for personal color analysis.

To enable a more systematic representation of personal color, particularly in East Asian, the Practical Color Coordinate System (PCCS) was developed by the Japan Color Research Institute in 1964 [3]. It provides a structured color space for describing colors in terms of hue, lightness, and saturation [20]. PCCS organizes colors into a 24-hue wheel and introduces the concept of tone as a composite attribute of lightness and saturation, allowing perceptual description such as 'pale', 'vivid', or 'dark'. The structure of the PCCS color wheel and its categories are shown in Fig. 1.

Personal Color Analysis. Personal color analysis is the process to identify the color palette that harmonizes best and worst with an individual's natural appearance, including skin tone, eye color, and hair color. By determining an individual's tonal characteristics, the analysis provides guidance for selecting suitable colors in areas such as fashion, cosmetics, and styling [1], [21]. Traditionally, personal color analysis is conducted through a draping procedure in which physical color fabrics are placed near an individual's face to observe how different colors interact with their facial features [1]. Through this process, trained experts evaluate subtle visual cues such as brightness, contrast, and harmony to determine the most suitable tonal category. However, this procedure largely relies on subjective judgment and professional experience, which may lead to inconsistencies across analysts. Moreover, the process requires specialized knowledge and in-person consultation, making it costly, time-consuming and difficult to scale. To address these challenges, various attempts have been made to quantitatively and systematically analyze and diagnose personal color for research and commercial applications. [22] studied quantitative model and proposed new personal color

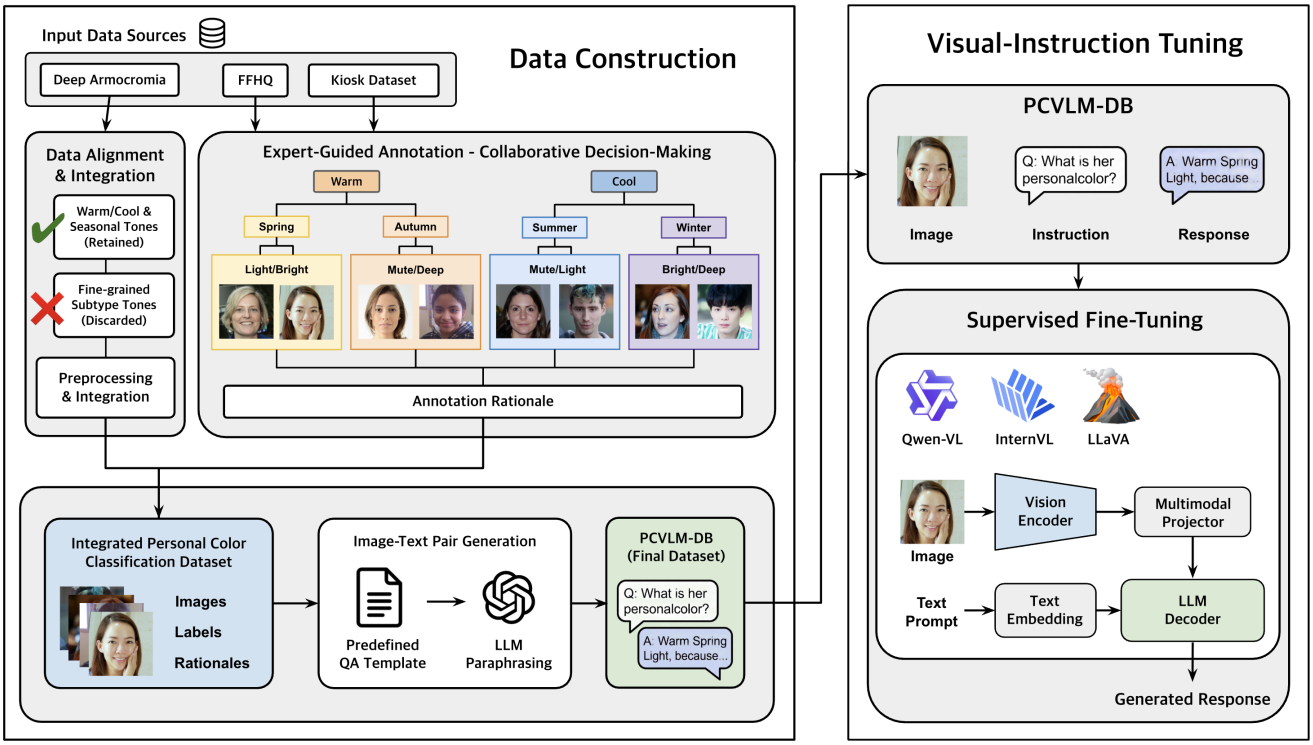


FIGURE 2. **Overview of the proposed pipeline. (Left) Data Construction: We curate PCVLM-DB by integrating public and proprietary images with expert-guided hierarchical annotations and LLM-based instruction generation. (Right) Visual-Instruction Tuning: Pre-trained VLMs (Qwen-VL, InternVL, LLaVA) are fine-tuned on PCVLM-DB to perform personal color classification and reasoning.**

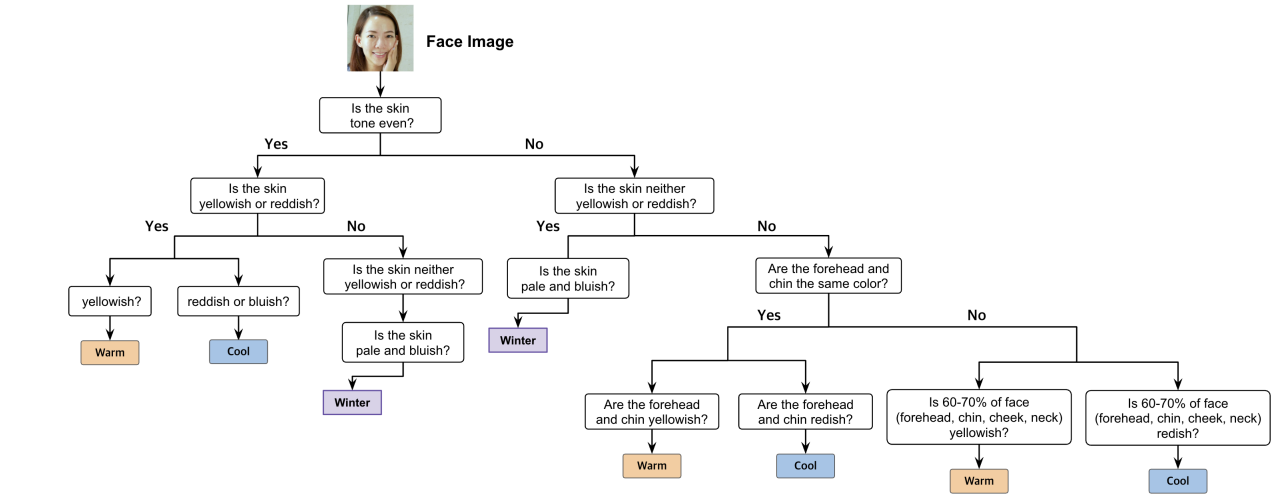
diagnosis model based on the PCCS, and [23] represented color palettes into the HSV color space to quantitatively analyze the characteristics of personal color.

On the other hand, several studies have attempted to address these challenges through algorithmic system design and user-interactive frameworks. [8] suggested decision-making algorithms for automatic color selection, and [7] proposed a mobile system for personal color analysis based on user-selected colors. [9] introduced Deep Armocromia, a deep learning-based approach for seasonal personal color classification, along with a novel dataset for personal color analysis. In addition, several open-source implementations have explored practical approaches to personal color analysis. [24] introduced a simple classification system with color filters, K-means algorithms, and convolutional neural networks. [25] approached the problem using dominant color extraction by transforming colors into HSV and Lab color space, while [26] employed facial part segmentation models to approximate traditional personal color theory. However, these approaches rely on relatively small datasets and show limited robustness under diverse environmental conditions, while also failing to provide interpretable reasoning for the personal color selection process.

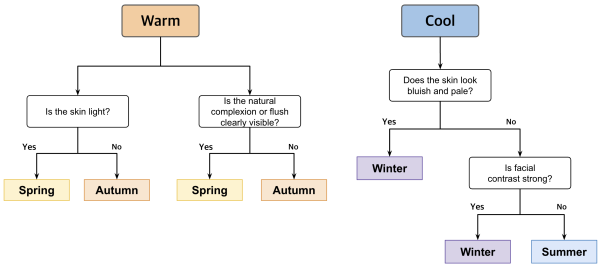
B. VISION-LANGUAGE MODELS

Vision Language Models (VLMs) [15], [27], [28] have recently shown remarkable success in multimodal understanding by extending the powerful reasoning capabilities of Large Language Models (LLMs) [29], [30] to the visual domain, effectively bridging computer vision and natural language processing through aligned visual and textual representations. In these models, visual features are extracted from images or videos, and projected into the language representation space to be processed jointly with text tokens, enabling the LLM decoder to perform complex multimodal reasoning and generation.

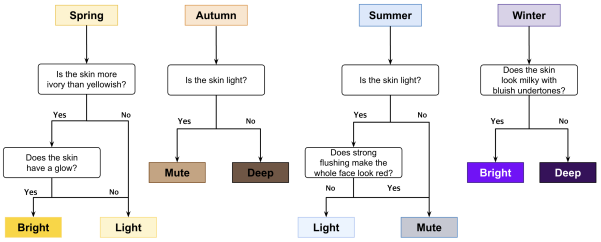
Recently, commercial multimodal systems such as ChatGPT [11], Gemini [12], and Claude [31] have demonstrated strong capabilities in visual understanding and multimodal reasoning across diverse real-world scenarios. This progress has accelerated the adoption of multimodal systems in practical applications across various industries, driving widespread AI transformation and shaping the modern AI era. For more accessible research and development, open-source VLMs such as Qwen-VL [14], [15], InternVL [27], [32], and the LLaVA series [28] have been introduced, providing scalable architectures for tasks including visual question answering, image captioning, image-grounded reasoning. These advances have further facilitated the application of VLMs to a variety of domain-specific tasks, including medical image analysis [18], [19], [33], scientific reasoning [34], and other specialized vision-language applications [16], [17], demonstrating profound proficiency in highly specialized diagnostic tasks. Therefore, the introduction of VLMs into the field of personal color analysis is strongly necessitated to leverage their powerful capabilities in multimodal comprehension and



(a) Warm vs Cool tone annotation



(b) Seasonal tone annotation



(c) Fine-grained subtype annotation

FIGURE 3. Hierarchical labeling process for personal color analysis. The procedure first determines the skin undertone (Warm, Cool) in (a), followed by seasonal classification (Spring, Summer, Autumn, Winter) in (b), and finally assigns a fine-grained subtype (Light, Bright, Muted, Deep) in (c). When the decision is ambiguous or cannot be clearly resolved from the flowchart, the final label is determined through discussion and consensus among annotators.

expert-level aesthetic judgments.

III. METHODOLOGY

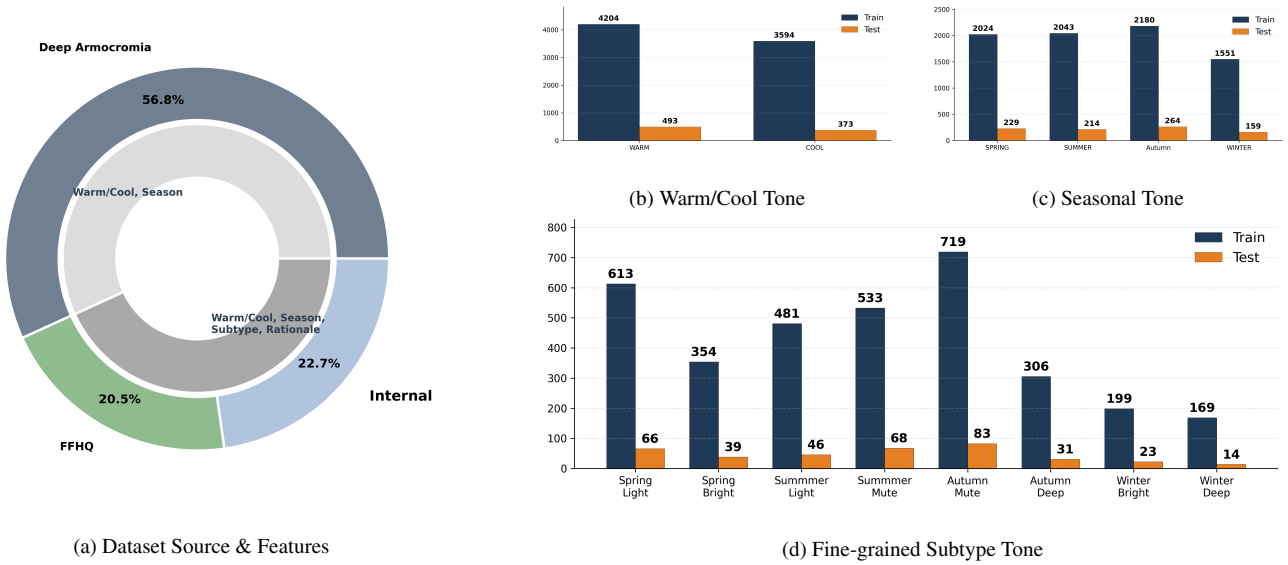
This section presents the systematic construction of the PCVLM-DB dataset and the subsequent visual instruction tuning framework for adapting pre-trained Vision-Language Models for specialized personal color analysis. The comprehensive workflow of our proposed method is illustrated in Fig. 2.

A. PCVLM-DB: MULTIMODAL PERSONAL COLOR DATASET

We construct PCVLM-DB by leveraging an existing public personal color dataset together with an internally curated dataset annotated by professional annotators. Our data curation pipeline consists of three stages: 1) data collection and expert-guided annotation, 2) data merging and refinement, and 3) converting the data into a vision-question answering format.

We first collected a public personal color classification dataset and combined it with an internally constructed personal color dataset. We partially adopt the public personal color dataset Deep Armocromia [9], which provides 4,920 facial images and categorized according to Armocromia Flow Theory. In addition, we construct an internal dataset consisting of 1,775 images sampled from FFHQ [35] and 1,969 proprietary frontal facial images acquired from an industrial imaging source (hereafter referred to as the Internal dataset). To ensure data quality, we implemented strict exclusion criteria, discarding images with heavy makeup, distorted lighting that masked natural skin color, or occlusions covering key diagnostic facial regions. All internal images were annotated by three trained annotators under the supervision of a domain expert, following a predefined decision protocol.

The personal color classification follows a hierarchical structure consisting of three levels: warm/cool tone, seasonal tone, and fine-grained subtype tone. At the top level, images are classified into warm or cool tones based on the dominance of yellow-based or pink/blue-based undertones. At the second level, warm tones are further divided into Spring (bright skin with visible vitality and peach/yellowish tones) and Autumn (earthy tones without prominent redness), while cool tones are divided into Summer (pale/bluish tint with subtle redness) and Winter (high contrast and intense cool brilliance). Finally, Spring includes Light (whitish skin) and Bright (luminous, transparent skin), Summer includes Mute (prominent surface redness or darker cool-tone) and Light



(a) Dataset Source & Features

(b) Warm/Cool Tone

(c) Seasonal Tone

(d) Fine-grained Subtype Tone

FIGURE 4. Comprehensive statistical and functional breakdown of the PCVLM-DB. (a) illustrates the dataset sources and integrated features. (b-d) show the hierarchical distribution of training and testing sets.

(fair skin without severe redness), Autumn includes Mute (brighter within Autumn) and Deep (deeply dark or blackish), and Winter includes Bright (clear, blue-based brilliance) and Deep (high contrast with dark features) subtypes.

To maintain annotation consistency, labels were determined through a multi-stage collaborative consensus following a predefined decision flow chart. Annotators collaboratively determined the final labels and provided textual rationales explaining the diagnostic reasoning behind each decision. Through this rigorous process, we achieved a highly reliable annotation outcome, with 57.4% of samples achieving perfect consensus (3:0) and 37.9% showing majority agreement (2:1), while 4.7% (1:1:1) exhibited complete disagreement. Specifically, among the disagreement cases, 38.6% (16.4% of the total) involved discrepancies at the fundamental Warm/Cool diagnosis, whereas 24.9% (10.6% of the total) and 36.5% (15.6% of the total) were attributed to differences at the seasonal tone and fine-grained subtype levels, respectively. The hierarchical label structure and the detailed decision flow are provided in Fig. 3.

Data Merging and Label Refinement. As the Deep Armochromia dataset follows different personal color theory and diagnostic criteria, we adopted only the binary warm/cool and seasonal tone labels, while excluding its fine-grained tone. All labels from different sources are normalized into a unified taxonomy, and translated into English. This preprocessing step ensures semantic alignment and enables joint training across public and internal data.

Image-Text Generation. Using the refined labels and annotated reasoning descriptions, we generate image-text pairs suitable for vision-language model training. Each image is associated with multiple single-turn question-answering pairs generated from predefined templates. The questions include classification queries targeting warm/cool tone, seasonal tone, or subtype tone, as well as queries requesting both classification results and explanatory reasoning. Answers are constructed in natural language based on the annotation rationale, allowing the model to learn both prediction and explanation. To enhance diversity, the test dataset is augmented by paraphrasing questions using LLMs while preserving their original meanings.

Overall, PCVLM-DB contains approximately 8,700 images and 30,000 textual question-answering pairs, consisting of 7,798 training images and 866 test images, where the training set includes 4,424 Deep Armocromia, 1,598 FFHQ, 1,776 Internal facial images, and the test set includes 496 Deep Armocromia, 177 FFHQ, and 193 Internal facial images. The comprehensive statistical and functional breakdown of PCVLM-DB is illustrated in Fig. 4. As shown in Fig. 4(a) , our dataset integrates both public and private sources to provide a rich set of features including professional rationales. In addition, the dataset follows a clear hierarchical structure. At the upper levels of classification, such as warm/cool and seasonal tones (Fig. 4(b), Fig. 4(c)), the dataset shows a stable and balanced distribution for each class. However, at the most detailed level, the fine-grained subtypes (Fig. 4(d)) exhibit imbalance, with winter subtypes appearing much less frequently than others.

B. VISUAL INSTRUCTION TUNING

Leveraging the PCVLM-DB dataset, we fully fine-tune multiple pre-trained VLMs via supervised fine-tuning (SFT). We adapt a diverse set of models covering various architectures, including Qwen3-VL [15], Qwen2.5-VL [14], InternVL3 [27], InternVL2.5 [32], and LLaVA-v1.5 [28]. As illustrated in Fig. 2, models consist of a vision encoder, a multimodal projector, and an LLM decoder. The vision encoder extracts visual features from input images, which are subsequently

TABLE 1. Comparative analysis of warm/cool and seasonal tone classification.

Model	Warm/Cool		Overall		Season				Overall	
	Warm	Cool	Accuracy	F1-Score	Spring	Summer	Autumn	Winter	Accuracy	F1-Score
Vision backbones + linear classifier										
Resnet-50	72.19	88.13	81.62	80.62	78.95	64.76	74.56	59.46	71.62	70.59
Resnet-101	74.83	91.32	84.59	83.69	82.46	71.43	75.44	62.16	75.14	74.74
Swin-T	75.50	88.58	83.24	82.42	77.19	70.48	66.67	70.27	71.35	71.81
Swin-S	63.58	91.32	80.00	78.28	82.46	73.33	72.81	70.27	75.68	76.65
Swin-B	80.13	87.21	84.32	83.74	83.33	67.62	78.95	72.97	76.49	76.43
ViT-T	82.78	85.39	84.32	83.87	73.68	74.29	80.70	59.46	74.59	74.03
ViT-S	59.60	95.89	81.08	78.86	82.46	83.81	56.14	59.46	72.43	72.52
ViT-B	71.52	89.50	82.16	81.09	80.70	74.29	65.79	64.86	72.70	72.66
ViT-L	62.91	89.95	78.92	77.19	62.28	89.52	22.81	70.27	58.65	57.80
Vision-Language Models (Zero-shot)										
ChatGPT-4o	41.58	64.61	51.49	-	26.20	41.59	19.32	30.82	38.75	-
Gemini3-Pro	64.50	56.03	60.85	-	40.17	35.98	50.00	47.17	43.42	-
LLaVA-v1.5-7B	99.68	0.32	56.84 ± 0.39	-	88.38	0.00	9.02	0.13	26.15 ± 8.07	-
InternVL3-8B	53.55	60.59	56.58 ± 17.69	-	31.70	54.58	29.70	17.61	34.16 ± 10.68	-
Qwen3-VL-8B	50.43	65.74	57.02 ± 12.84	-	25.68	27.66	38.41	54.72	35.27 ± 8.38	-
Vision-Language Models (SFT)										
LLaVA-v1.5-7B	88.44	36.93	66.17 ± 0.18	61.40 ± 0.21	87.25	41.31	19.17	20.25	42.84 ± 0.68	40.14 ± 0.33
InternVL2.5-8B	77.89	59.36	69.91 ± 0.31	68.69 ± 0.39	62.45	44.02	53.64	64.03	55.50 ± 1.14	55.36 ± 0.50
InternVL3-8B	54.93	89.60	69.86 ± 0.45	69.80 ± 0.83	29.17	76.73	55.45	82.26	58.68 ± 0.53	57.58 ± 0.63
Qwen2.5-VL-3B	74.00	72.12	73.19 ± 0.21	75.02 ± 0.08	61.31	63.08	60.98	70.06	63.26 ± 0.26	65.46 ± 0.30
Qwen3-VL-4B	86.29	85.04	85.75 ± 0.47	86.04 ± 0.41	77.21	80.47	75.53	78.49	77.74 ± 0.70	78.13 ± 0.37
Qwen2.5-VL-7B	90.47	90.24	90.37 ± 0.21	90.22 ± 0.19	78.86	88.32	84.24	86.04	84.16 ± 0.25	84.53 ± 0.21
Qwen3-VL-8B	93.67	87.77	$\mathbf{91.13 \pm 0.33}$	$\mathbf{91.43 \pm 0.24}$	88.21	84.67	84.47	86.79	$\mathbf{85.94 \pm 0.40}$	$\mathbf{86.58 \pm 0.34}$

projected into the language embedding space via a multimodal projector and integrated with text tokens for processing by the LLM decoder. All models are optimized using the standard autoregressive next-token prediction objective. Given an input facial image x_{img}, a textual instruction x_{txt}, and the target response $Y = \{y_1, y_2, \ldots, y_T\}$, the models minimize the negative log-likelihood:

$$\mathcal{L}_{\text{SFT}} = -\sum_{t=1}^{T} \log P_\theta(y_t \mid x_{img}, x_{txt}, y_{<t}) \tag{1}$$

where t denotes the index of the current token in a sequence of total length T, $y_{<t}$ represents the sequence of tokens generated prior to step t, and θ represents the trainable parameters. Through this SFT process, these general-purpose models are adapted into domain-specific experts, which we collectively refer to as PC-VLMs (Personal Color Vision-Language Models), enabling them to effectively associate visual chromatic features with the diagnostic reasoning and criteria used in personal color analysis.

IV. EXPERIMENTS

A. EXPERIMENTAL SETUP

We evaluate model performance using a multiple-choice question (MCQ) format. Each question is presented as a natural-language query, and models are required to select an answer corresponding to predefined class categories along with textual rationales for their decisions. To ensure consistency across models, all VLMs are fully fine-tuned using the same instruction format. Training is conducted for three epochs on a single NVIDIA H200 GPU using the AdamW [36] optimizer, bfloat16 precision, and DeepSpeed ZeRO-3 [37] settings. As a vision-only baseline, we additionally train classifier models based on pretrained timm backbones [38]. These models use separate classification heads for Warm/Cool, Season, and fine-grained prediction.

B. EXPERIMENTAL RESULTS

We compare PC-VLMs with conventional vision-classifiers and zero-shot vision-language models. The quantitative results for warm/cool and seasonal tone classification are summarized in Table 1, and the performance for fine-grained subtype classification is shown in Table 2. We also present a qualitative analysis to validate the models' ability to provide interpretable diagnostic reasoning and visually consistent predictions aligned with human judgment.

Comparative Analysis. We first evaluate the performance of PC-VLMs against a diverse set of baselines, including conventional vision-only classifiers and commercial models. Table 1 presents a comprehensive comparison across the fundamental warm/cool and seasonal tone classification tasks. As shown in the results, our fine-tuned Qwen3-VL-8B establishes a highly effective baseline for the personal color classification task, recording 91.13% in warm/cool classification and 85.94% in seasonal tone classification, with F1-scores reaching 91.43% and 86.58%, respectively. This significantly outperforms conventional vision-only classifier models, which exhibit reasonable performance in warm/cool tasks but heavily struggle with more complex seasonal classifications. More notably, PC-VLMs significantly outperform

TABLE 2. Detailed performance comparison on fine-grained subtypes, including overall accuracy and F1 score.

Model	Warm				Cool				Overall	
	Spring		Autumn		Summer		Winter		Accuracy	F1-Score
	Light	Bright	Mute	Deep	Mute	Light	Bright	Deep		
Vision backbones + linear classifier										
Resnet-50	60.24	83.87	60.87	51.47	42.42	84.62	26.09	57.14	57.84	56.25
Resnet-101	69.88	87.10	63.04	57.35	39.39	89.74	52.17	35.71	62.43	61.87
Swin-T	72.29	70.97	52.17	55.88	56.06	79.49	56.52	50.00	62.70	61.77
Swin-S	75.90	80.65	76.09	39.71	54.55	76.92	65.22	35.71	63.78	63.90
Swin-B	69.88	77.42	76.09	51.47	57.58	64.10	60.87	42.86	63.51	63.22
ViT-T	63.86	77.42	82.61	39.71	57.58	82.05	60.87	35.71	62.43	63.24
ViT-S	65.06	80.65	39.13	47.06	50.00	92.31	39.13	42.86	57.57	56.50
ViT-B	68.67	80.65	50.00	60.29	36.36	89.74	69.57	28.57	60.81	59.21
ViT-L	62.65	48.39	30.43	26.47	51.52	92.31	69.57	21.43	50.81	48.38
Vision-Language Models (Zero-shot)										
ChatGPT-4o	6.06	5.13	8.43	16.14	27.94	32.61	21.74	0.00	15.41	-
Gemini3-Pro	12.12	17.95	34.94	35.48	16.18	28.26	39.13	42.86	25.41	-
LLaVA-v1.5-7B	79.70	12.82	0.00	5.16	0.00	0.00	0.00	1.43	16.05 ± 7.60	-
InternVL3-8B	22.12	11.79	9.88	38.06	35.59	43.04	15.65	11.43	23.89 ± 12.63	-
Qwen3-VL-8B	1.52	34.36	7.95	52.26	0.59	38.70	48.70	42.86	19.62 ± 3.69	-
Vision-Language Models (SFT)										
LLaVA-v1.5-7B	59.39	34.87	8.92	0.00	46.18	34.35	26.96	0.00	30.70 ± 0.62	25.36 ± 0.23
InternVL2.5-8B	36.67	48.21	68.92	14.19	0.00	55.22	37.39	10.00	37.84 ± 0.60	30.56 ± 1.32
InternVL3-8B	36.06	24.10	67.71	78.71	53.24	81.30	38.26	68.57	55.62 ± 0.52	53.81 ± 0.93
Qwen2.5-VL-3B	33.64	74.87	47.23	80.65	55.88	71.74	60.00	35.71	55.51 ± 0.24	60.24 ± 0.47
Qwen3-VL-4B	63.64	72.31	66.51	87.74	64.41	79.57	50.43	40.00	67.62 ± 0.67	66.01 ± 1.51
Qwen2.5-VL-7B	59.09	73.85	67.47	78.71	71.18	79.13	56.52	58.57	68.70 ± 0.77	68.39 ± 0.82
Qwen3-VL-8B	69.39	80.00	70.60	74.19	65.29	67.83	65.22	70.00	$\mathbf{70.00 \pm 0.72}$	$\mathbf{70.47 \pm 0.88}$

commercial general-purpose models such as ChatGPT-4o [10] and Gemini3-Pro [13]. The best-performing model of PC-VLMs surpasses these commercial models by more than 30% in warm/cool classification and 40% in seasonal tone classification.

We further evaluate the performance on fine-grained subtype classification across 8 tone categories, as summarized in Table 2. This task is more challenging than warm/cool and seasonal tone classification due to the subtle tonal distinctions between subtypes. As reported in this table, our best-performing model achieves an overall accuracy of 70.00% and a Macro F1-score of 70.47%. Specifically, it outperforms commercial zero-shot models by 44.59% in overall accuracy and further achieves a 6.22% improvement over the most stable vision-only classifier. In addition, the model exhibits more balanced performance across subtype classes. While conventional vision-only and zero-shot models show extreme performance fluctuations, with accuracy variations of at least 29.41% and 30.74% across subtypes, respectively, the best-performing PC-VLM shows a much smaller variation of only 14.78%. These results demonstrate that our model provides more reliable and stable predictions for personal color classification while achieving superior overall performance.

Among the evaluated VLM architectures, the Qwen-VL series consistently achieves the best performance across all levels of personal color classification. In particular, Qwen3-VL-8B outperforms other VLM families such as InternVL and LLaVA. Additionally, larger-scale models are more effective than smaller model for personal color analysis. To further validate the robustness of PC-VLMs, we additionally evaluate the models on each dataset, including the Armocromia dataset, FFHQ dataset, and our Internal dataset. The results show similar performance trends across datasets, where the Qwen-VL series achieves the best performance among the evaluated models. Performance comparisons for each specific dataset is provided in Appendix A.

Qualitative Results. We present qualitative analysis on representative personal color classification results from the PCVLM-DB test set. Fig. 5 shows example visual question answering outputs by our PC-VLMs across various personal color categories and data sources. As illustrated in the figure, the generated responses are not limited to simple classification but include comprehensive textual rationales that justify the diagnostic decisions. The responses also reflect hierarchical reasoning by identifying the warm/cool tone, seasonal type, and fine-grained subtype. Although the Deep Armocromia dataset does not provide rationales or subtype annotations, the models are still able to generate plausible



Image
(Deep Armocromia)

Question: Identify the most suitable personal color for this individual and provide a rationale for your choice.

Ground Truth: COOL WINTER

PC-VLM Response: Based on the visual features shown in the image, this person is classified as **COOL WINTER DEEP**. They fall under the cool tone group with a winter seasonal tone and a deep subtype. The reasoning behind this classification includes: The entire face has a bluish tone with a reddish hue on the cheeks, the skin is pale and bright with strong contrast, and dark colors with low brightness suit well.



Image
(FFHQ)

Question: What personal color classification does this person have, and why?

Ground Truth: WARM SPRING LIGHT

PC-VLM Response: This person has a warm-toned personal color with a spring seasonal tone and a light subtype. The final personal color classification is **WARM SPRING LIGHT**. This conclusion is based on the following observed characteristics: The face has a yellowish tone, with bright yellowish skin, a healthy complexion, and spring warm colors suit well. The skin does not have much radiance and is not very white.

FIGURE 5. **Qualitative examples of personal color classification produced by PC-VLM. For each example, the model predicts the personal color category and provides reasoning based on visual chromatic cues.**

explanations and infer fine-grained subtype characteristics based on visual cues. Additional examples are provided in Appendix B.

V. DISCUSSION

In this work, we propose PC-VLMs, a domain-specific vision-language approach to personal color analysis. By constructing PCVLM-DB with expert-guided rationales, visual features are aligned with structured chromatic descriptions, enabling PC-VLMs to perform personal color classification with interpretable explanations. Our experimental results show that PC-VLMs significantly outperform both conventional vision-only classifiers and existing commercial multimodal models, demonstrating that VLMs can effectively classify and explain sensory and aesthetic attributes that rely on complex visual harmony.

Despite high accuracy of PC-VLMs, we occasionally observe discrepancies in fine-grained subtype estimation. As illustrated in Fig. 6, the model shows a gap between its visual perception and interpretation relative to human aesthetic judgment. Fig. 6(a) highlights the confusion between Spring light and bright subtypes. While the model reasons that the subject shows "sheen" on the face, the expert assessment indicates that the skin actually lacks a significant gloss. This shows a difference in how the model and human observers interpret "gloss". In addition, in Fig. 6(b), the model misclassified a Cool Winter Deep subject as Warm Spring Light. The model's rationale identified a "yellowish tone" and "bright yellowish skin" as primary indicators. However, the expert assessment describes an overall bluish undertone with localized redness on the cheeks, noting a pale complexion with strong contrast. This suggests that the model tends to struggle with accurately capturing structural contrast and the subtle boundary between yellowish and pinkish hues in pale skin. Given that the subject's face simultaneously exhibits both yellowish and reddish pale skin, such ambiguity naturally poses a challenge for precise classification. This highlights the importance of refining the model's sensitivity to these minute tonal shifts to reduce the gap between AI and human perception for the personal color analysis.

In addition, despite these promising outcomes, the generated responses follow a relatively fixed and rigid pattern due to reliance on supervised fine-tuning. To address this, future work will investigate preference optimization or reinforcement learning approaches to better align the model with diverse expert consensus and general user preferences, thereby encouraging more flexible and adaptive diagnostic reasoning for ambiguous or borderline cases. Furthermore, to apply our research to real-world applications, we will explore extending our framework to downstream tasks, including personalized fashion and cosmetic recommendations.

Taken together, these observations clarify the intended positioning of PC-VLMs. Given the inherently subjective nature of personal color analysis and the residual gap between model



Ground Truth: WARM SPRING LIGHT

Rationale: Yellowish undertone across the face; bright yellow skin with slight redness. Suits whitish Spring-Warm colors; skin lacks significant gloss, and is not excessively pale.

PC-VLM Response: Based on the visual features shown in the image, this person is classified as **WARM SPRING BRIGHT**. They fall under the warm tone group with a spring seasonal tone and a bright subtype. The reasoning behind this classification includes: Overall, there is no redness, and the area around the cheeks has an orange tone. The eye skin is on the lighter side among warm tones, and the skin is white and has a sheen.

(a) Spring Light misclassified as Bright due to perceived sheen.



Ground Truth: COOL WINTER DEEP

Rationale: Overall bluish undertone with localized redness on the cheeks. The complexion is pale and bright, exhibiting strong contrast between facial features and skin. Best suited for dark, low-value colors.

PC-VLM Response: Based on the visual features shown in the image, this person is classified as **WARM SPRING LIGHT**. They fall under the warm tone group with a spring seasonal tone and a light subtype. The reasoning behind this classification includes: The face has a yellowish tone, with bright yellowish skin, a healthy complexion, and spring warm colors suit well. The skin does not have much radiance and is not very white.

(b) Winter Deep misclassified as Spring Light due to misinterpretation of undertone and contrast.

FIGURE 6. Representative failure cases in fine-grained subtype estimation. The model's rationales are contrasted with expert annotations to highlight discrepancies between pixel-level sensitivity and human aesthetic judgment.

predictions and expert judgment in fine-grained or ambiguous cases, PC-VLMs are not intended to replace professional consultants. Instead, we envision PC-VLMs as a decision-support framework that operates alongside human experts: providing scalable, interpretable preliminary analyses for broader user populations, while deferring to expert review for borderline or high-stakes cases. In professional contexts, the model's hierarchical predictions and textual rationales can serve as an auxiliary reference that helps stylists communicate their reasoning to clients and maintain consistency across analysts, rather than as an autonomous diagnostic system.

VI. CONCLUSION

Personal color analysis requires sensory and aesthetic judgment of visual harmony and chromatic characteristics. This process is inherently subjective and difficult to standardize, as it relies on the expertise of individual stylists to interpret the interaction between skin and colors. To address these challenges, we introduce Personal Color Vision-Language Models (PC-VLMs), a scalable and interpretable domain-specific vision-language framework that can support expert-level personal color analysis as a decision-support tool. We constructed PCVLM-DB, the first comprehensive VQA dataset for personal color analysis, with hierarchical classification labels and professional diagnostic rationales. Through visual instruction tuning, we demonstrated that general-purpose VLMs can be effectively adapted to capture perceptual and aesthetic judgments involved in personal color analysis. Experimental results show that PC-VLMs significantly outperform both conventional vision-only classification models and proprietary zero-shot multimodal models, achieving higher classification accuracy and more stable performance across personal color categories while providing interpretable reasoning. These results highlight the potential of VLMs for both classification and interpretable reasoning in aesthetic and sensory visual domains, positioning PC-VLMs as a complement to, rather than a replacement for, expert consultation. Despite these advances, PC-VLMs still face challenges in capturing subtle tonal variations and fully aligning with human aesthetic judgment, particularly in fine-grained and ambiguous cases, which further underscores the value of keeping human experts in the loop. Future work will explore preference alignment and reinforcement learning techniques to encourage more flexible and adaptive reasoning for personal color analysis. We believe this work provides a foundation for future research on multimodal systems for fashion, beauty, and personalized styling applications that require perceptual and aesthetic judgment.

APPENDIX A DATASET-WISE PERFORMANCE ANALYSIS

Performance comparisons for each specific dataset are provided below.

TABLE A.1. Classification Performance for Warm/Cool and Seasonal Tones — Armocromia dataset

Model	Warm/Cool			Season				
	Warm	Cool	Overall	Spring	Summer	Autumn	Winter	Overall
LLaVA-v1.5-7B	96.57	23.87	64.03 ± 0.34	96.94	25.00	18.53	18.36	39.40 ± 0.71
InternVL2.5-8B	67.74	60.45	64.48 ± 0.66	62.32	35.6	36.4	71.64	52.14 ± 1.38
InternVL3-8B	40.80	89.19	62.46 ± 0.46	18.87	69.40	38.67	84.75	51.25 ± 0.41
Qwen2.5-VL-3B	75.62	67.48	71.98 ± 0.25	63.55	48.60	59.20	71.48	61.17 ± 0.37
Qwen3-VL-4B	84.89	86.22	85.48 ± 0.32	76.77	79.20	71.60	83.77	77.72 ± 1.04
Qwen2.5-VL-7B	92.63	92.61	92.62 ± 0.27	82.26	90.60	89.47	90.82	88.23 ± 0.18
Qwen3-VL-8B	95.99	91.80	$\mathbf{94.11 \pm 0.17}$	90.48	90.60	89.33	91.15	$\mathbf{90.32 \pm 0.14}$

TABLE A.2. Classification Performance for Warm/Cool and Seasonal Tones — FFHQ dataset

Model	Warm/Cool			Season				
	Warm	Cool	Overall	Spring	Summer	Autumn	Winter	Overall
LLaVA-v1.5-7B	91.46	42.43	70.96 ± 0.51	90.18	25	18.53	18.36	49.60 ± 0.93
InternVL2.5-8B	91.65	58.65	77.85 ± 0.74	77.19	35.6	36.4	71.64	65.31 ± 1.53
InternVL3-8B	67.57	95.68	79.32 ± 1.81	45.96	69.4	38.67	84.75	73.56 ± 1.46
Qwen2.5-VL-3B	86.41	87.57	86.89 ± 0.25	77.19	48.6	59.2	71.48	79.66 ± 0.00
Qwen3-VL-4B	89.51	82.7	86.67 ± 1.17	83.86	79.2	71.6	83.77	81.02 ± 0.51
Qwen2.5-VL-7B	89.32	85.14	87.57 ± 0.00	78.25	90.6	89.47	90.82	81.69 ± 0.51
Qwen3-VL-8B	95.53	79.19	$\mathbf{88.70 \pm 0.4}$	92.98	90.6	89.33	91.15	$\mathbf{83.62 \pm 0.4}$

TABLE A.3. Classification Performance for Warm/Cool and Seasonal Tones — Kiosk Dataset

Model	Warm/Cool			Season				
	Warm	Cool	Overall	Spring	Summer	Autumn	Winter	Overall
LLaVA-v1.5-7B	66.55	68.31	67.25 ± 1.00	58.75	68	23.48	32.67	45.49 ± 0.68
InternVL2.5-8B	89.66	56.88	76.58 ± 0.68	37.5	53.43	78.26	48.00	55.13 ± 0.70
InternVL3-8B	77.07	84.94	80.21 ± 0.85	35.83	77.14	83.48	82.67	64.15 ± 0.93
Qwen2.5-VL-3B	59.14	70.65	63.73 ± 0.63	36.67	70.57	78.26	80.00	53.58 ± 0.79
Qwen3-VL-4B	86.72	83.9	85.60 ± 0.43	70.42	84.29	84.78	75.33	75.54 ± 0.43
Qwen2.5-VL-7B	86.38	88.31	87.15 ± 0.68	70.83	89.14	82.61	86.67	75.96 ± 0.59
Qwen3-VL-8B	86.55	84.42	$\mathbf{85.70 \pm 1.19}$	76.67	84.29	83.48	83.33	$\mathbf{76.79 \pm 1.69}$

TABLE A.4. Detailed Accuracy for Fine-grained Sub-types — FFHQ Dataset

Model	Warm				Cool				Overall
	Spring		Autumn		Summer		Winter		
	Light	Bright	Mute	Deep	Mute	Light	Bright	Deep	
LLaVA-v1.5-7B	65.48	37.50	0.00	0.00	26.25	30.21	32.89	0.00	26.10 ± 0.74
InternVL2.5-8B	43.81	52.22	60.74	18.95	0.00	43.33	45.26	12.73	38.64 ± 2.48
InternVL3-8B	31.43	26.11	62.22	94.74	67.00	59.17	45.26	74.55	57.85 ± 0.64
Qwen2.5-VL-3B	27.62	81.11	51.85	100.00	46.00	91.67	72.63	43.64	66.54 ± 0.25
Qwen3-VL-4B	49.52	75.56	66.67	95.79	49.00	77.50	61.05	50.91	67.46 ± 0.64
Qwen2.5-VL-7B	28.57	80.00	62.96	86.32	55.00	87.50	68.42	74.55	68.59 ± 1.42
Qwen3-VL-8B	47.62	86.67	64.44	84.21	50.00	74.17	78.95	72.73	$\mathbf{70.85 \pm 0.64}$

TABLE A.5. Detailed Accuracy for Fine-grained Sub-types — Kiosk Dataset

Model	Warm				Cool				Overall
	Spring		Autumn		Summer		Winter		
	Light	Bright	Mute	Deep	Mute	Light	Bright	Deep	
LLaVA-v1.5-7B	56.67	0.00	12.05	0.00	53.65	40.91	0.00	0.00	34.92 ± 0.79
InternVL2.5-8B	33.33	0.00	72.86	6.67	0.00	68.18	0.00	0.00	37.10 ± 1.35
InternVL3-8B	38.22	0.00	70.36	53.33	47.5	72.73	5.00	46.67	53.58 ± 0.46
Qwen2.5-VL-3B	36.44	0.00	45.00	50.00	60.00	50.00	0.00	6.67	45.39 ± 0.59
Qwen3-VL-4B	70.67	33.33	66.43	75.00	70.83	81.82	0.00	0.00	67.88 ± 0.97
Qwen2.5-VL-7B	73.33	0.00	69.64	66.67	77.92	70.00	0.00	0.00	68.81 ± 0.43
Qwen3-VL-8B	79.56	0.00	73.57	58.33	71.67	60.91	0.00	60.00	$\mathbf{69.22 \pm 1.25}$

APPENDIX B ADDITIONAL QUALITATIVE EXAMPLES

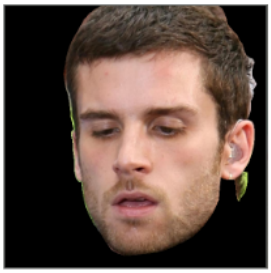


Image
(Deep Armocromia)

Question: What personal color classification does this person have, and why?

Ground Truth: WARM SPRING

PC-VLM Response: The image suggests a warm tone combined with a spring seasonal category and a light subtype, resulting in the personal color classification **WARM SPRING LIGHT**. This assessment is supported by these observed features: The skin has yellow base, is relatively bright, and is typically classified as yellow rather than white.



Image
(FFHQ)

Question: Determine the best personal color category for this person and briefly explain your reasoning.

Ground Truth: WARM SPRING BRIGHT

PC-VLM Response: Based on the visual features shown in the image, this person is classified as **WARM SPRING BRIGHT**. They fall under the warm tone group with a spring seasonal tone and a bright subtype. The reasoning behind this classification includes: Overall, there is no redness, and the area around the cheeks has an orange tone. The eye skin is on the lighter side among warm tones, and the skin is white and has a sheen.



Image
(Kiosk Dataset)

Question: Would you please diagnose which personal color best fits this individual and share the reasons for your decision?

Ground Truth: WARM AUTUMN MUTE

PC-VLM Response: Based on the visual features shown in the image, this person is classified as **WARM AUTUMN MUTE**. They fall under the warm tone group with a fall seasonal tone and a mute subtype. The reasoning behind this classification includes: The skin is close to brown, the skin is relatively dark, it is not black and dark, and when muted tones are applied to the face, the face appears brighter.



Image
(Deep Armocromia)

Question: Determine the best personal color category for this person and briefly explain your reasoning.

Ground Truth: WARM AUTUMN

PC-VLM Response: The image suggests a warm tone combined with a fall seasonal category and a deep subtype, resulting in the personal color classification **WARM AUTUMN DEEP**. This assessment is supported by these observed features: Warm brown skin tone, on the darker side of warm tones, skin tone is uniform, very dark skin tone.

FIGURE B.1. Qualitative examples of Warm tone personal color classification produced by PC-VLM.



Image (FFHQ)

Question: Determine the best personal color category for this person and briefly explain your reasoning.

Ground Truth: COOL SUMMER MUTE

PC-VLM Response: The image suggests a cool tone combined with a summer seasonal category and a mute subtype, resulting in the personal color classification **COOL SUMMER MUTE**. This assessment is supported by these observed features: The skin is generally on the red side, the face is red so there is no pale feeling, there is a lot of red energy in the face, and the skin is not bright.



Image (FFHQ)

Question: Identify the most suitable personal color for this individual and provide a rationale for your choice.

Ground Truth: COOL SUMMER LIGHT

PC-VLM Response: Based on the visual features shown in this image, this person is classified as **COOL SUMMER LIGHT**. They fall under the cool tone group with a summer seasonal tone and a light subtype. The reasoning behind this classification includes: The cheeks are very red, and there is a reddish hue on the forehead and chin. The face is not pale white, and the overall contrast is not hight. Light tones with a hint of white match better than muted tones with a hint of gray. The skin tone is on the brighter side.



Image (FFHQ)

Question: Would you please diagnose which personal color best fits this individual and share the reasons for your decision?

Ground Truth: COOL WINTER BRIGHT

PC-VLM Response: Based on the visual features shown in the image, this person is classified as **COOL WINTER BRIGHT**. They fall under the cool tone group with a winter seasonal tone and a bright subtype. The reasoning behind this classification includes: Light ivory skin tone, with dark eye brows and irises creating a strong contrast with the skin color, pale white complexion, bright winter colors in primary shade suit better than dark colors



Image (Kiosk Dataset)

Question: Would you please diagnose which personal color best fits this individual and share the reasons for your decision?

Ground Truth: COOL WINTER DEEP

PC-VLM Response: Based on the visual features shown in the image, this person is classified as **COOL WINTER DEEP**. They fall under the cool tone group with a winter seasonal tone and a deep subtype. The reasoning behind this classification includes: The face is white with a blue undertone, the face is pale and has a sense of contrast. The hair is very dark, and dark colors mixed with black suit well.

FIGURE B.2. Qualitative examples of Cool tone personal color classification produced by PC-VLM.

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REFERENCES

[1] C. Jackson, *Color me beautiful: Discover your natural beauty through the colors that make you look great and feel fabulous*. Ballantine Books, 1987.
[2] D. Fujii and J. von Alten, *Color with Style*. Graphic-Sha Publishing Company, 1992. [Online]. Available: https://books.google.co.kr/books?id=h5YGAAAACAAJ
[3] J. C. R. Institute, "Practical color coordinate system (pccs)," https://sikiken.co.jp/, 1964, [Online; webpage].
[4] S. Herndon, "Ultimate guide to color analysis: How to determine your best colors for clothing and makeup," https://radiantlydressed.com/color-analysis/, 2024, [Online; webpage post].
[5] J.-Y. Lee, "A study on the color arrangement of personal color & beauty and fashion," *Journal of the Korean Society of Cosmetology*, vol. 29, no. 2, pp. 485–498, 2023.
[6] A. H. Munsell, *A color notation*. Munsell color company, 1919.
[7] H. R. Hong and Y. I. Kim, "A mobile application for personal colour analysis," *Cogent Business & Management*, 2019.
[8] T. Withurat, W. Sripen, J. Pattanasukkul, W. Wongsim, S. Jeeratanyasakul, and T. Siriborvornratanakul, "Personal color analysis using color space algorithm," *Advances in Computational Intelligence*, vol. 4, no. 2, pp. 1–10, 2024.
[9] L. Stacchio, M. Paolanti, F. Spigarelli, and E. Frontoni, "Deep armocromia: A novel dataset for face seasonal color analysis and classification," in *European Conference on Computer Vision*. Springer, 2024, pp. 352–367.
[10] A. Hurst, A. Lerer, A. P. Goucher, A. Perelman, A. Ramesh, A. Clark, A. Ostrow, A. Welihinda, A. Hayes, A. Radford *et al.*, "Gpt-4o system card," *arXiv preprint arXiv:2410.21276*, 2024.
[11] A. Singh, A. Fry, A. Perelman, A. Tart, A. Ganesh, A. El-Kishky, A. McLaughlin, A. Low, A. Ostrow, A. Ananthram *et al.*, "Openai gpt-5 system card," *arXiv preprint arXiv:2601.03267*, 2025, model documentation available at https://arxiv.org/pdf/2601.03267 and https://cdn.openai.com/pdf/3a4153c8-c748-4b71-8e31-aecbde944f8d/oai_5_2_system-card.pdf.
[12] G. Comanici, E. Bieber, M. Schaekermann, I. Pasupat, N. Sachdeva, I. Dhillon, M. Blistein, O. Ram, D. Zhang, E. Rosen *et al.*, "Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities," *arXiv preprint arXiv:2507.06261*, 2025.
[13] Google DeepMind, "A new era of intelligence with Gemini 3," https://blog.google/products-and-platforms/products/gemini/gemini-3/, Nov 2025, google Official Blog.
[14] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge *et al.*, "Qwen2.5-vl technical report," *arXiv preprint arXiv:2502.13923*, 2025.
[15] S. Bai, Y. Cai, R. Chen, K. Chen, X. Chen, Z. Cheng, L. Deng, W. Ding, C. Gao, C. Ge *et al.*, "Qwen3-vl technical report," *arXiv preprint arXiv:2511.21631*, 2025.
[16] C. Cui, Y. Ma, X. Cao, W. Ye, Y. Zhou, K. Liang, J. Chen, J. Lu, Z. Yang, K.-D. Liao *et al.*, "A survey on multimodal large language models for autonomous driving," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 2024, pp. 958–979.
[17] K. Kawaharazuka, J. Oh, J. Yamada, I. Posner, and Y. Zhu, "Vision-language-action models for robotics: A review towards real-world applications," *IEEE Access*, 2025.
[18] C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon, and J. Gao, "Llava-med: Training a large language-and-vision assistant for biomedicine in one day," *Advances in Neural Information Processing Systems*, vol. 36, pp. 28 541–28 564, 2023.
[19] J. Wu, Y. Wang, Z. Zhong, W. Liao, N. Trayanova, Z. Jiao, and H. X. Bai, "Vision-language foundation model for 3d medical imaging," *npj Artificial Intelligence*, vol. 1, no. 1, p. 17, 2025.
[20] S. gui Lee, "A study on the color arrangement education model using pccs(practical color co-ordinate system) color system," Master's Thesis, Kookmin University Graduate School of Education, Seoul, 2009.
[21] P. Sirisayan, "Personal color analysis based on color harmony for skin tone," 2022.
[22] Y.-S. Jung, "A study on the quantitative diagnosis model of personal color," *Journal of Convergence for Information Technology*, vol. 11, no. 11, pp. 277–287, 2021.
[23] T. D. Cassidy, "Personal colour analysis, consumer colour preferences and colour forecasting for the fashion and textile industries," *Colour: Design & Creativity*, vol. 1, no. 1, p. 6, 2007.
[24] slmteruto, "Cai (color artificial intelligence)," https://github.com/slmteruto/CAI?tab=readme-ov-file, 2020, [Online; webpage; github].
[25] starbucksdolcelatte, "Show me the color," https://github.com/starbucksdolcelatte/ShowMeTheColor, 2019, [Online; webpage; github].
[26] PSY222, "Colorinsight (personal color detection model)," https://github.com/PSY222/Colorinsight, 2023, [Online; webpage; github].
[27] J. Zhu, W. Wang, Z. Chen, Z. Liu, S. Ye, L. Gu, H. Tian, Y. Duan, W. Su, J. Shao *et al.*, "Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models," *arXiv preprint arXiv:2504.10479*, 2025.
[28] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," *Advances in neural information processing systems*, vol. 36, pp. 34 892–34 916, 2023.
[29] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *Advances in neural information processing systems*, vol. 33, pp. 1877–1901, 2020.
[30] W. X. Zhao, K. Zhou, J. Li, T. Tang, X. Wang, Y. Hou, Y. Min, B. Zhang, J. Zhang, Z. Dong *et al.*, "A survey of large language models," *arXiv preprint arXiv:2303.18223*, vol. 1, no. 2, pp. 1–124, 2023.
[31] Anthropic, "Claude 3.5 Sonnet Model Card," https://www.anthropic.com/news/claude-3-5-sonnet, 2024.
[32] Z. Chen, W. Wang, Y. Cao, Y. Liu, Z. Gao, E. Cui, J. Zhu, S. Ye, H. Tian, Z. Liu *et al.*, "Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling," *arXiv preprint arXiv:2412.05271*, 2024.
[33] Z. Zeng, Z. Zhuo, X. Jia, E. Zhang, J. Wu, J. Zhang, Y. Wang, C. H. Low, J. Jiang, Z. Zheng *et al.*, "Surgvlm: A large vision-language model and systematic evaluation benchmark for surgical intelligence," *arXiv preprint arXiv:2506.02555*, 2025.
[34] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. Li, Y. Wu *et al.*, "Deepseekmath: Pushing the limits of mathematical reasoning in open language models," *arXiv preprint arXiv:2402.03300*, 2024.
[35] T. Karras, S. Laine, and T. Aila, "A style-based generator architecture for generative adversarial networks," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 4401–4410.
[36] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *arXiv preprint arXiv:1711.05101*, 2017.
[37] S. Rajbhandari, J. Rasley, O. Ruwase, and Y. He, "Zero: Memory optimizations toward training trillion parameter models," in *SC20: international conference for high performance computing, networking, storage and analysis*. IEEE, 2020, pp. 1–16.
[38] R. Wightman, "Pytorch image models," https://github.com/rwightman/pytorch-image-models, 2019.



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